

Ex. No. 10	Use Ghidra to disassemble and analyze the malware code
Date	

Project Objectives

- To provide a hands-on guide to disassembling and analyzing malware with Ghidra.
- To enable readers to recognize common malware functionalities, such as persistence mechanisms, anti-analysis techniques, and network communications.
- To equip users with the skills to interpret low-level assembly code and understand high-level behaviours.

GitHub Repository Components

A GitHub repository for this project could include:

1. README.md

- Overview: Brief description of Ghidra, its installation, and how it is used for malware analysis.
- Requirements: List of requirements, including Ghidra, a VM setup, and any sample binaries (preferably benign samples or detailed steps to safely acquire controlled malware samples).
- Project Outline: A breakdown of the tutorial steps, linking to various sections or files within the repository.

2. Tutorial Steps (Markdown or Jupyter Notebooks)

- Step-by-step instructions, broken down into digestible sections, such as:
 - Environment Setup: Instructions on setting up a virtualized and isolated environment.
 - Initial Analysis: How to load a binary into Ghidra, perform auto-analysis, and identify entry points.
 - Function Analysis: Methods for identifying and analyzing critical functions, decompiling code, and using Ghidra's cross-referencing features.
 - String and Import Analysis: How to locate and interpret strings and imports related to malicious behavior.

- Advanced Techniques: Dynamic analysis, control flow graphs, custom scripting with Ghidra's Python or Java capabilities, and obfuscation detection.

3. Example Scripts (Python or Java)

- Scripts to automate repetitive analysis tasks, such as:
 - Extracting and listing all string references.
 - Identifying network-related functions or common indicators of malware.
 - Labeling functions and data segments based on known patterns (e.g., file system manipulation, registry edits).
- Script Examples: Python or Java code snippets demonstrating API usage in Ghidra (using Ghidra's Headless Analyzer, for example, for automation).

4. Sample Data (Benign Binary or Hex Dump)

- Due to safety and ethical concerns, include only safe, benign binaries or a hex dump example rather than live malware. This can allow users to learn and practice without the risk of executing malicious code.
- Sample binaries designed to exhibit typical malware-like behaviors, such as:
 - Executing system calls.
 - Writing to files.
 - Sending network data.
- Alternatively, a dummy binary generator script that creates binaries with similar structures to malware but without harmful payloads.

3. Documentation and Comments

- Include explanations of the code within each script, detailing what each function does.
- Inline comments in disassembly views, providing readers with insights into assembly commands, control flow structures, and the decompiled code.

6. Report Template

- A reporting template for documenting findings, which could be in Markdown or PDF format. The template should cover:
 - Summary of Analysis: Brief description of the suspected malware.
 - Function Analysis: Summary of the key functions, their purpose, and the findings.

- Behavioral Indicators: Summary of malware behaviors, such as persistence, file system manipulation, or network communications.
- Recommendations: Suggestions on further actions, such as containment, eradication, or further analysis.

Example Repository Structure

plaintext

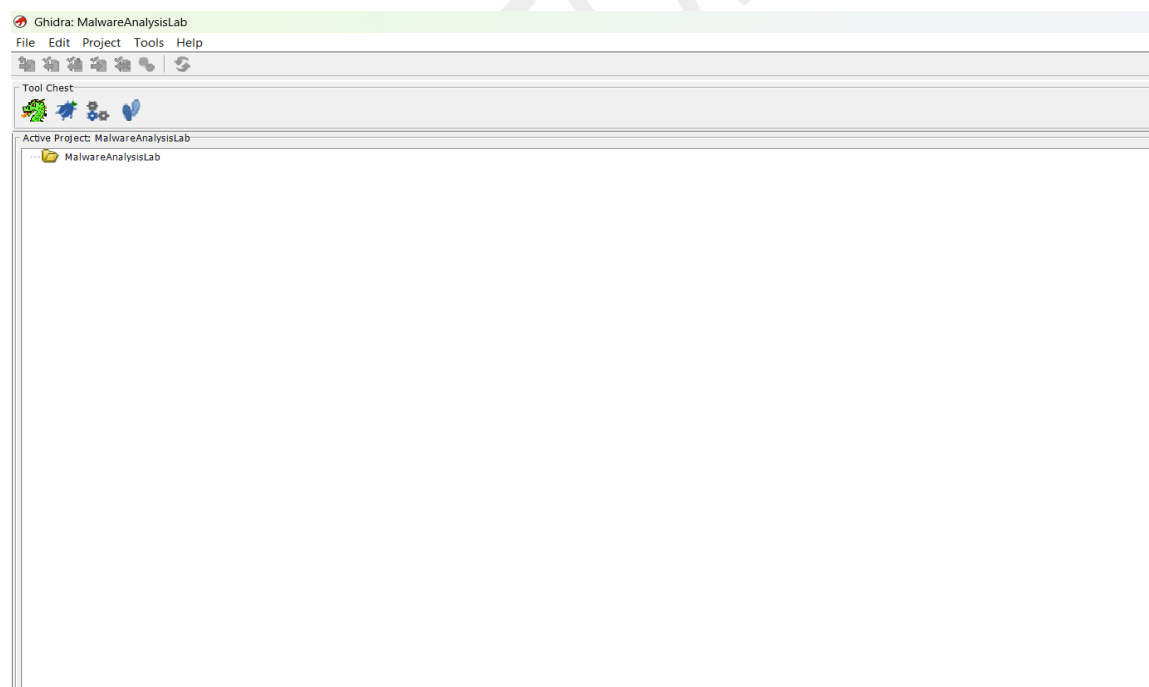
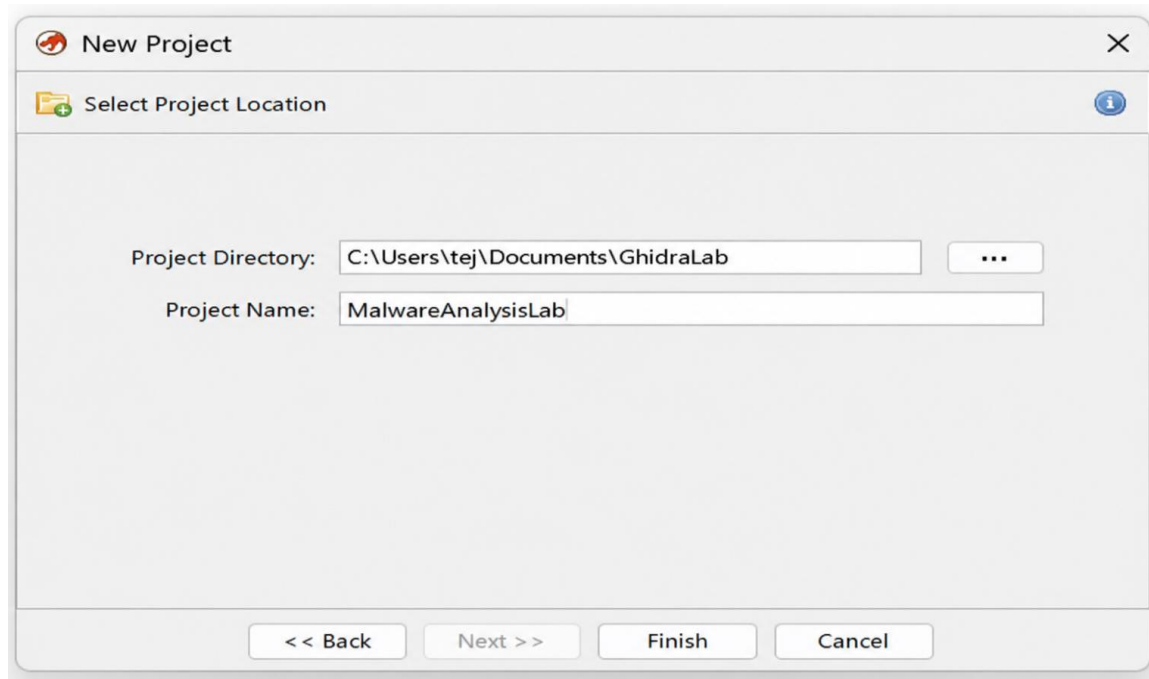
ghidra-malware-analysis-project/

```
|— README.md
|— docs/
|   |— Tutorial_Step1.md
|   |— Tutorial_Step2.md
|   |— Tutorial_StepN.md
|— scripts/
|   |— extract_strings.py
|   |— network_analysis.py
|   |— label_functions.py
|— samples/
|   |— benign_sample1.bin
|   |— benign_sample2.hex
|— templates/
|   |— analysis_report_template.md
|— screenshots/
|   |— function_graph_example.png
|   |— decompiler_example.png
```

Project Development and Future Work

- Expand Scripts: Continue adding Ghidra scripts to automate complex analysis tasks.
- Community Contributions: Encourage community members to contribute additional benign samples, analysis scripts, and documentation improvements.
- Advanced Topics: Include guides for advanced techniques, such as unpacking obfuscated code, identifying encryption routines, and using the P-Code emulator for runtime analysis.

Output Screenshots:



Defined Strings (CodeBrowser: MalwareAnalysisLab/ping.exe)

Location	String Value	String Representation	Data Type
140000000	.MZ	"MZ"	char[2]
140000100	PE	"PE"	char[4]
140000200	.text	".text"	char[8]
140000230	.fothk	".fothk"	char[8]
140000238	.rdata	".rdata"	char[8]
140000280	.data	".data"	char[8]
1400002A8	.pdata	".pdata"	char[8]
1400002D0	.rsrc	".rsrc"	char[8]
1400002F8	.reloc	".reloc"	char[8]
1400054C8	OutputEncoding	"OutputEncoding"	unicode
1400054A8	Unicode	"Unicode"	unicode
1400054B8	Ans	"Ans"	unicode
140005508	UTF8	"UTF8"	unicode
140005518	UTF-8	"UTF-8"	unicode
140005540	SVCSTMT(CurrentControlSet\Services\Tcpip\Parameters	"SVCSTMT(CurrentControlSet\Services\Tcpip\Parameters"	unicode
140005548	DefaultTTL	"DefaultTTL"	unicode
1400055C0	IPv4	"IPv4"	unicode
1400055D0	IPv6	"IPv6"	unicode
140005600	RSOS	"RSOS"	ds
140005708	ping.pdb	"ping.pdb"	ds
14000582C	.gchcont	".gchcont"	ds
140005850	.rdata	".rdata"	ds
140005860	.rdata\$voltnid	".rdata\$voltnid"	ds
140005890	.data	".data"	ds
140005910	.pdata	".pdata"	ds
14000592	.exit	".exit"	ds
1400059A	._get_oshandle	"._get_oshandle"	ds
1400059Ac	._wscicmp	"._wscicmp"	ds
1400059B8	.flush	".flush"	ds
1400059C2	.swctype	".swctype"	ds
1400059C8	.wscstul	".wscstul"	ds
1400059D8	._setmode	"._setmode"	ds
1400059E4	._write	"._write"	ds
1400059E8	._flush	"._flush"	ds
1400059F8	.wscchr	".wscchr"	ds
140006002	.fgetpos	".fgetpos"	ds
14000600C	.fwprint	".fwprint"	ds
140006018	._XcpFilter	"._XcpFilter"	ds
140006028	._ping_exit	"._ping_exit"	ds
140006034	._getmainargs	"._getmainargs"	ds
140006046	._set_app_type	"._set_app_type"	ds
140006058	._exit	"._exit"	ds
140006060	._ceit	"._ceit"	ds
14000606A	._setusermatherr	"._setusermatherr"	ds
140006074	._initterm	"._initterm"	ds
14000608A	._C_specific_handler	"._C_specific_handler"	ds
140006092	._fmode	"._fmode"	ds
14000609C	._rtimeofday	"._rtimeofday"	ds

Filter:

CodeBrowser: MalwareAnalysisLab/ping.exe

File Edit Analysis BSim Graph Navigation Search Select Tools Window Hel

Program Trees

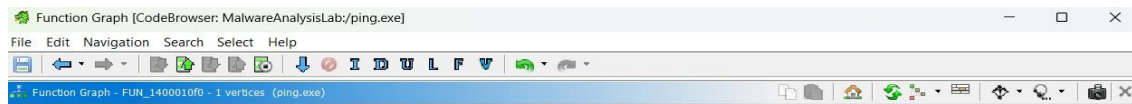
- ping.exe
 - Headers
 - .text
 - .fothk
 - .rdata
 - .data
 - .pdata
 - .rsrc
 - .reloc
 - Debug Data
 - tdb

Program Tree X

Symbol Tree

- API-MS-WIN-CORE-PROCESS-THREADS-L1-1-0.DLL
- API-MS-WIN-CORE-PROFILE-L1-1-0.DLL
- API-MS-WIN-CORE-REGISTRY-L1-1-0.DLL
- API-MS-WIN-CORE-RTLSUPPORT-L1-1-0.DLL
- API-MS-WIN-CORE-STRING-L1-1-0.DLL
- API-MS-WIN-CORE-SYNCH-L1-2-0.DLL
- API-MS-WIN-CORE-SYSINFO-L1-1-0.DLL
- IPHLPAPI.DLL
- MSVCRT.DLL
- NTDLL.DLL
- WS2_32.DLL
- Exports

Filter:



```

1400010f0 - FUN_1400010f0
undefined __fastcall FUN_1400010f0(void)
undefined <UNASSIGNED> <RETURN>
undefined8 Stack[-0x18]:8 local_18
FUN_1400010f0
...10f0 SUB RSP,0x38
...10f4 MOV EAX,dword ptr [DAT_1400076...]
...10fa LEA RS,[DAT_140007158]
...1101 MOV R9D,dword ptr [DAT_1400076...]
...1108 LEA RDX,[DAT_140007150]
...110f MOV dword ptr [DAT_140007164]...
...1115 LEA RCX,[DAT_140007148]
...111c LEA RAX,[DAT_140007164]
...1123 MOV qword ptr [RSP + local_18...]
...1128 CALL qword ptr [MSVCRT.DLL:...]
...112e MOV dword ptr [DAT_14000714c]...
...1134 ADD RSP,0x38
...1138 RET

```



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The malware sample was successfully disassembled and analyzed using Ghidra, including its functions, strings, imports, and control flow.

The analysis helped identify potential malware behaviors and understand the program's functionality through assembly code and decompiled output.